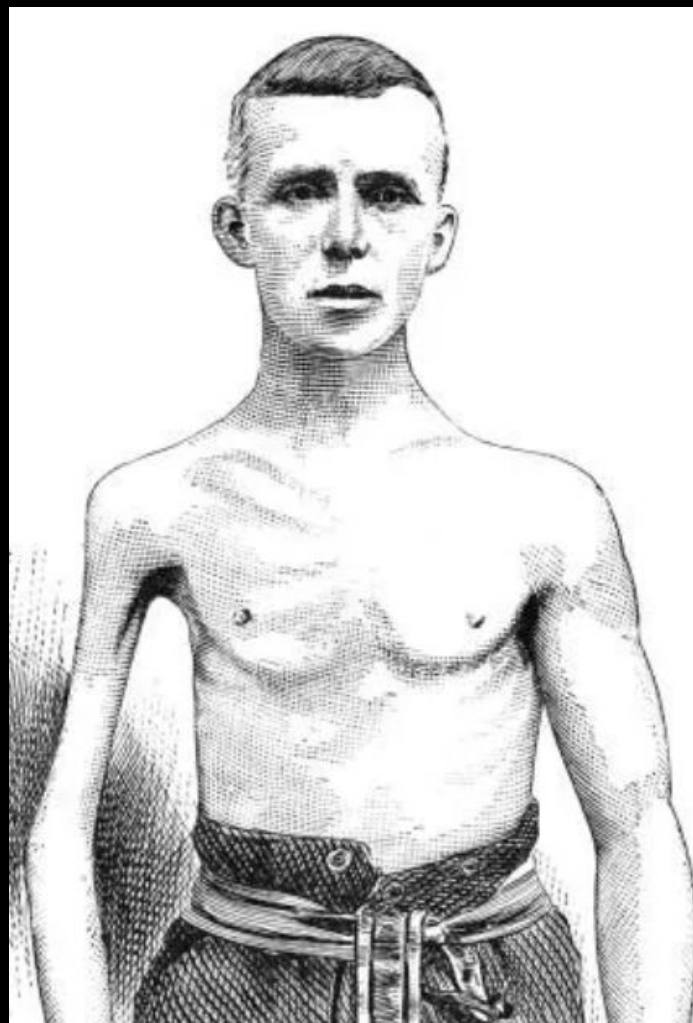


MyoTens V2

**Annie, Zack, Tyler, and
Colton**

Muscle atrophy

Cleveland
Clinic
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EMG Circuit:

$$G = 1 + \frac{49.4 \text{ k}\Omega}{R_g} \Rightarrow 2 = 1 + \frac{49.4 \text{ k}\Omega}{R_g} \Rightarrow R_g = 5.49 \text{ k}\Omega \text{ (theoretical)}$$

$$R_g = 5.1 \text{ k}\Omega \text{ (experimental)}$$

High pass: gain = 5

$$f_c = \frac{1}{2\pi R_i C} \Rightarrow 50\text{Hz} = \frac{1}{2\pi R_i (0.1 \times 10^{-6} \text{F})} \Rightarrow R_i = 31.83 \text{ k}\Omega \text{ (theoretical)}$$

$$R_i = 22 \text{ k}\Omega \text{ in series with } 10 \text{ k}\Omega \text{ (experimental)}$$

$$\text{Gain} = -\frac{R_f}{R_i} \Rightarrow 5 = -\frac{R_f}{31.83 \text{ k}\Omega} \Rightarrow R_f = 318.3 \text{ k}\Omega \text{ (theoretical)}$$

$$R_f = 220 \text{ k}\Omega \text{ in series with } 100 \text{ k}\Omega \text{ (experimental)}$$

$$\text{Capacitor} = 0.1 \mu\text{F}$$

Low pass: gain = 5

$$f_c = \frac{1}{2\pi R_i C} \Rightarrow 150\text{Hz} = \frac{1}{2\pi R_i (0.1 \times 10^{-6} \text{F})} \Rightarrow R_i = 10.6 \text{ k}\Omega \text{ (theoretical)}$$

$$R_i = 10 \text{ k}\Omega \text{ (experimental)}$$

$$\text{Gain} = -\frac{R_f}{R_i} \Rightarrow 5 = -\frac{R_f}{10.6 \text{ k}\Omega} \Rightarrow R_f = 106 \text{ k}\Omega \text{ (theoretical)}$$

$$R_f = 100 \text{ k}\Omega \text{ in parallel with } 5.1 \text{ k}\Omega$$

$$\text{Capacitor} = 0.1 \mu\text{F}$$

Inverting op-amp:

$$\text{Gain} = \frac{V_o}{V_i} = -\frac{R_f}{R_i} \Rightarrow 2 = -\frac{20}{10}$$

$$R_f = 100 \text{ k}\Omega \text{ in series and } R_i = 13 \text{ k}\Omega$$

Total Gain ~ 100 (hypothetical)

```
1 const int redLEDPin = 2;
2 const int greenLEDPin = 4;
3 const int emgPin = 34;
4
5 const int EMG_THRESHOLD = 500;
6 const int SAMPLE_WINDOW = 10;
7 const int SAMPLE_DELAY = 5;
8 const int GRACE_PERIOD = 250; // in milliseconds
9
10 bool muscleActive = false;
11 unsigned long lastActiveTime = 0;
12
13 void setup() {
14     Serial.begin(115200);
15     pinMode(redLEDPin, OUTPUT);
16     pinMode(greenLEDPin, OUTPUT);
17
18     digitalWrite(redLEDPin, HIGH);
19     digitalWrite(greenLEDPin, LOW);
20 }
21
22 void loop() {
23     double sumOfSquares = 0;
24
25     for (int i = 0; i < SAMPLE_WINDOW; i++) {
26         int emgValue = analogRead(emgPin);
27         sumOfSquares += (double)(emgValue * emgValue);
28         delay(SAMPLE_DELAY);
29     }
30
31     double rms = sqrt(sumOfSquares / SAMPLE_WINDOW);
32
33     // Plot RMS and activity status
34     Serial.print(rms);
35     Serial.print("\t");
36     Serial.println(muscleActive ? 1 : 0);
37
38     if (rms > EMG_THRESHOLD) {
39         lastActiveTime = millis();
40         if (!muscleActive) {
41             muscleActive = true;
42             digitalWrite(greenLEDPin, HIGH);
43             digitalWrite(redLEDPin, LOW);
44         }
45     } else {
46         if (muscleActive && millis() - lastActiveTime > GRACE_PERIOD) {
47             muscleActive = false;
48             digitalWrite(greenLEDPin, LOW);
49             digitalWrite(redLEDPin, HIGH);
50         }
51     }
52
53     delay(50);
54 }
```

Why MyoTens?

- **Reduces Muscle Atrophy**
- **Affordable**
- **Easy to Use**

THANK YOU